

Gaming the Two Numbers They Control: Quantization, Weight-Cut Bunching, and the Limits of Strength in 3.9 Million Powerlifting Results

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Abstract—A powerlifting competitor controls exactly two numbers: the weight loaded on the bar and their bodyweight at weigh-in. We ask whether these choices leave measurable statistical patterns, using ~ 3.94 million competition results from the public-domain OpenPowerlifting database. We report four findings, each answered with a course-appropriate test and led by effect size (at $n \approx 3.9\text{M}$ almost everything is “significant”). (i) *Quantization*: 96.2% of 13.4M attempt loads sit exactly on the 2.5 kg plate grid; a generalized likelihood-ratio test (GLRT) whose null parameter lies on the boundary is calibrated by a parametric bootstrap. (ii) *Weight-cut bunching*: bodyweight piles up just below every IPF men’s class limit (de-heaped $\log(\text{below/above}) = +1.92$ at 83 kg); a Cattaneo–Jansson–Ma density-discontinuity test rejects at six of the seven limits (surviving Holm and Benjamini–Hochberg); the lone exception is the 120 kg boundary of the open-ended 120+ class, where the incentive to cut is weakest. A non-limit control shows no bunching, and placebos show no bunching-direction false positives. (iii) *Allometry*: strength scales with bodyweight differently by sex ($b = 0.75$ men, 0.51 women vs. the isometric $2/3$; a $\text{Sex} \times \log(\text{BW})$ interaction is highly significant and persists on common support). (iv) *Prediction*: bodyweight, sex, equipment, and age predict the total ($R^2 = 0.70$, random forest, leakage-safe grouped CV). Together, the two numbers competitors control leave clear statistical fingerprints, discreteness on the bar and strategic mass below the limit, while the scaling and extreme-value analyses find no evidence of a population strength ceiling. Code and data: <https://github.com/adirelm/opl-stat-theory>.

I. INTRODUCTION

Powerlifting comprises three lifts (squat, bench press, dead-lift), each with three attempts, contested within bodyweight classes. Two features make it useful for studying strategic behavior. First, the implements are *discrete*: plates come in fixed denominations, so the load a competitor can place on the bar is not a continuous choice. Second, eligibility is governed by a *threshold*: a lifter who weighs even slightly above a class limit is bumped into a heavier, stronger field. We focus on the two numbers a competitor records: the load chosen for each attempt, and the bodyweight at weigh-in. Both interact with a sharp administrative rule. We ask whether these two choices leave detectable statistical signatures, and what $\sim 3.94\text{M}$ results reveal about the limits of human strength.

Related work. Manipulation of a “running variable” around an administrative threshold produces *bunching*, a standard tool in public economics [1] that is classically tested with the McCrary density-discontinuity test [2] and its modern, bias-corrected local-polynomial form due to Cattaneo, Jansson, and

Ma [3]. Rapid pre-weigh-in weight cutting is well documented in weight-class sports [4]. On the physiological side, the square–cube law predicts that strength, governed by muscle cross-sectional area, scales with bodyweight as $\propto \text{BW}^{2/3}$. Extreme-value theory has been used to estimate the ultimate limits of athletic performance from record data [5]. On OpenPowerlifting specifically, recent work studies *scoring and scaling* systems [6]; to our knowledge, no formal manipulation test of weight-cut bunching has been applied to this dataset.

Contribution. We organize the paper around the two recorded choices, load and weigh-in bodyweight, and use both halves of the course, classical inference and machine learning. Our contributions are: (1) a rounded-likelihood *quantization* GLRT for the plate grid, with a parametric-bootstrap calibration that respects the boundary nature of the null; (2) to our knowledge the first formal density-discontinuity test of weight-cut *bunching* on OpenPowerlifting, checked using a non-limit control, placebo cutoffs, de-heaping, and multiple-testing corrections; (3) a sex-interaction *allometry* model that tests the scaling exponent against the isometric $2/3$; and (4) a leakage-safe strength-*prediction* model. Throughout we lead with effect sizes and confidence intervals rather than p -values, as explained next.

Methodological stance. At $n \approx 3.9\text{M}$ the power to detect any non-zero effect is effectively one, so a p -value answers a question we do not care about (“is the effect exactly zero?”) rather than the one we do (“is it large?”). We therefore report effect sizes with 95% confidence intervals as the primary evidence, use p -values only as a secondary screen, and apply multiple-testing corrections to every pre-declared family: all four (Bonferroni, Holm, Šidák, Benjamini–Hochberg) for the H2 limits, and Holm for H3. Because the same lifter appears in many rows, naive standard errors understate uncertainty (pseudo-replication); the formal tests therefore use one row per lifter (with heteroskedasticity-robust HC3 errors), the deduplication unit declared per hypothesis.

II. RESULTS

All intervals are 95%. H1 uses attempt loads; the cross-sectional tests use one row per lifter (to respect independence), and where the class scheme matters the modern kg-only era (2014+). Headline effect sizes use all rows.

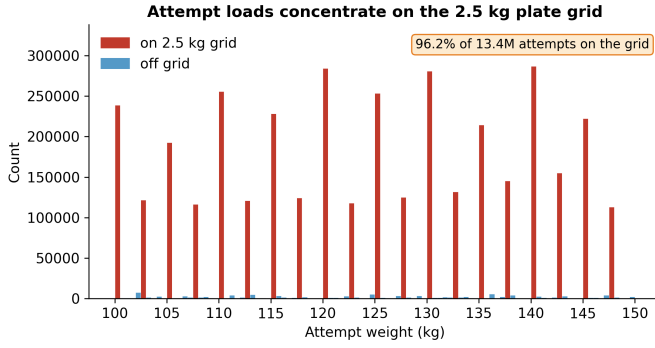


Fig. 1. H1: attempt loads concentrate on the 2.5 kg plate grid.

A. Attempt loads are quantized to the 2.5 kg grid (H1)

Of 13,397,702 valid attempt loads, **96.20%** (CI [96.19, 96.21]) fall exactly on the 2.5 kg plate grid (Fig. 1); the smallest standard plate is 1.25 kg, loaded on both sides, so 2.5 kg is the minimum increment. To test this formally we restrict to the kg-native (0.5 kg) lattice and model the within-cell position as a mixture that places weight π on the grid point and $1 - \pi$ on a uniform background. The GLRT of $\pi = 0$ versus $\pi > 0$ has its null parameter on the boundary of $[0, 1]$, so Wilks' theorem fails: the null distribution of $2 \ln \Lambda$ is the 50:50 mixture $\frac{1}{2} \chi_0^2 + \frac{1}{2} \chi_1^2$ of Chernoff and Self-Liang [7], [8]. Our parametric bootstrap closely matches this theory (point mass 0.51 vs. the theoretical 0.50; calibrated 95% cutoff 2.59 vs. the theoretical 2.71, both well below the naive χ_1^2 value of 3.84). The observed statistic, $2 \ln \Lambda = 3.9 \times 10^7$, exceeds every bootstrap draw ($p < 3.4 \times 10^{-4}$), and a χ^2 goodness-of-fit against a uniform within-cell distribution strongly rejects it (5.0×10^7 , $df = 4$). The 3.8% off-grid loads are largely a units artifact: of the 2.1% that fall off the 0.5 kg lattice entirely, 82% are round pound loads (e.g., 102.06 kg = 225 lb), themselves quantized on a different grid.

B. Bodyweight bunches below class limits (H2)

Bodyweight piles up just below real class limits (Fig. 2). On de-heaped data (exact round/half-kilogram weigh-ins removed, so the signal is not mere digit preference), the log-ratio of just-below to just-above counts is positive at all seven IPF men's limits and strongest at 83 kg ($+1.92 \pm 0.04$, a $6.8 \times$ asymmetry; Table I, Fig. 3); a non-limit control at 91 kg is flat-to-slightly-negative (-0.21). The formal Cattaneo–Jansson–Ma density-discontinuity test [3] (one random meet per lifter, de-heaped, 2014+) rejects at six of the seven limits with the density dropping at the cutoff ($t < 0$), surviving both Holm and Benjamini–Hochberg correction; the lone exception is the 120 kg cutoff ($t = -1.6$, $p = 0.11$), adjacent to the unbounded 120+ class where the incentive to cut is weakest. The control shows no excess below (a small *opposite*-direction discontinuity), and no placebo shows the bunching pattern. The effect is robust across equipment (raw $+2.14$, equipped $+1.51$) and concentrated in the tested majority ($+1.93$; the smaller untested subset reverses to -0.30 , consistent with weight-

TABLE I
DE-HEAPED $\log(\text{BELOW}/\text{ABOVE})$ AT EACH IPF MEN'S LIMIT (ALL ROWS; COUNTS OVER THE 0.5 KG WINDOWS JUST BELOW/ABOVE EACH LIMIT).
LAST ROW IS A NON-LIMIT CONTROL.

Limit (kg)	log-ratio	$\pm$	$\times$ asymmetry
59	+0.99	0.04	2.7
66	+1.18	0.03	3.2
74	+1.00	0.03	2.7
83	+1.92	0.04	6.8
93	+1.65	0.04	5.2
105	+1.17	0.05	3.2
120	+0.74	0.06	2.1
91 (control)	-0.21	0.03	0.8

cutting being driven by the tested IPF-scheme federations that dominate this population), and it persists on the clean kg-only subset ($+1.80$).

Robustness and falsification. The result survives four checks. (i) *De-heaping*: removing exact round/half-kilogram weigh-ins leaves the asymmetry intact, so the signal is not digit preference and is consistent with weight-cut manipulation. (ii) *Control*: the non-limit point at 91 kg shows no excess below. (iii) *Placebos*: cutoffs placed between real limits show no bunching-direction discontinuity after correction. (iv) *Subgroups and correction*: the effect holds within raw, equipped, and tested subsets, and the formal density test rejects at six of the seven limits even under the strictest (Bonferroni) correction.

C. Allometric scaling differs by sex (H3)

Fitting $\log(\text{Total}) = a + b \log(\text{BW})$ (Fig. 4), the exponent is $b = 0.75$ (CI [0.74, 0.75]) for men and 0.51 ([0.50, 0.51]) for women, with heteroskedasticity-robust (HC3) standard errors on one row per lifter. A two-sided Wald test rejects the isometric $b = 2/3$ for both sexes, and in *opposite* directions (men $z = +46.8$, i.e. $b > 2/3$; women $z = -64.1$, i.e. $b < 2/3$). A pooled $\text{Sex} \times \log(\text{BW})$ interaction estimates the men-minus-women slope gap at $+0.24$ ($p \ll 10^{-3}$). The gap also *grows* on the common bodyweight support (0.84 men vs. 0.38 women), so it is not an artifact of the sexes occupying different weight ranges. For men, the Pearson correlation of $\log \text{BW}$ and $\log \text{Total}$ is 0.60 (Fisher-transform CI [0.600, 0.603]; Spearman 0.58); it is lower for women. The message is not that women are weaker, but that the *estimated scaling* differs; we leave the mechanism to future work.

D. Strength is predictable from basic variables (H4)

Predicting the total from controllable/basic variables (bodyweight, sex, equipment, age), on one random meet per lifter subsampled to a seeded 200,000 lifters for tractability, with every split grouped by lifter and all preprocessing fit inside each training fold, a random forest reaches $R^2 = 0.70$ (RMSE 86 kg) versus 0.61 (RMSE 98 kg) for linear regression (Fig. 5); permutation importance is dominated by sex and bodyweight, and continuous-feature VIFs are ≈ 1 (no collinearity concern). A logistic “made-weight” classifier built from *non-bodyweight*

Bodyweight bunches just below a real class limit, not at a non-limit control

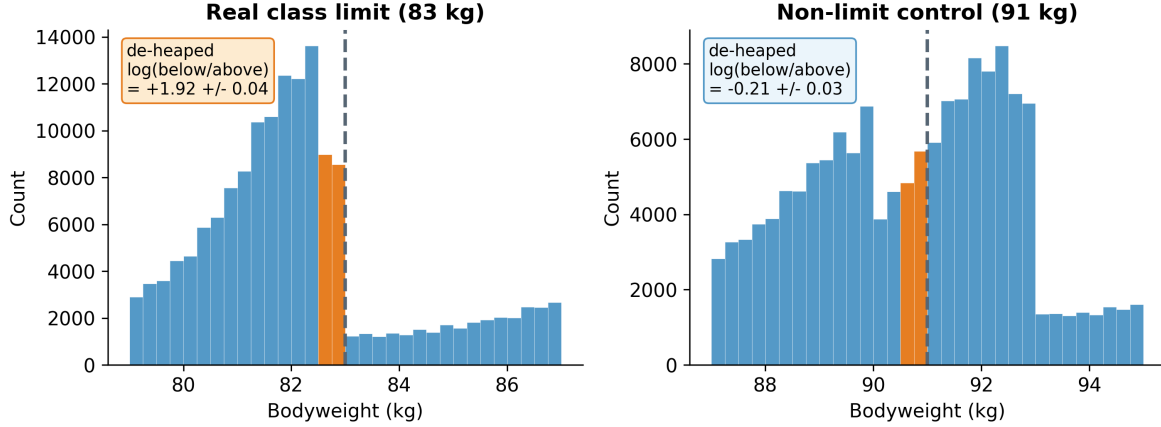


Fig. 2. H2: bunching just below a real limit (83 kg) but not at a non-limit control (91 kg); round-number weigh-ins removed.

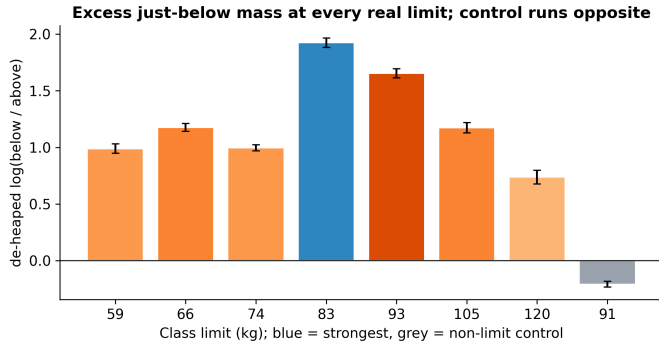


Fig. 3. H2: excess just-below mass at every real limit; the control runs opposite.

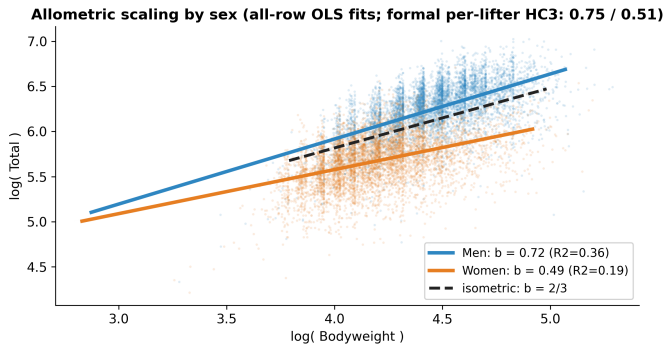


Fig. 4. H3: allometric scaling by sex vs. the isometric 2/3. Plotted lines are all-row OLS fits; the headline per-lifter HC3 exponents are 0.75 (men) / 0.51 (women).

features (age, tested status, era, equipment; the Dots score is excluded because it is a function of total and bodyweight) reaches AUC = 0.56 on $n = 89,882$ lifters (PR-AUC 0.84 against a 0.82 no-skill baseline, i.e. the below-limit share; balanced accuracy 0.53). The signal is weak, as expected: cutting is driven by bodyweight relative to the limit, which

defines the label and which these features omit.

E. Supporting analyses

Distribution structure. The apparent two-component shape of raw totals is just sex: by BIC and a bootstrap likelihood-ratio test, raw Total prefers two Gaussian components ($\Delta\text{BIC} = 1.4 \times 10^3$) but the sex-normalized Dots score prefers one (Fig. 6); the two-component shape largely disappears after Dots normalization. **Tested vs. untested.** Untested competitions show higher Dots than tested ones (medians 344 vs. 318, Mann–Whitney $p \rightarrow 0$; a parametric Welch test agrees, $|t| = 128$; rank-biserial -0.20), consistent with (though not proof of) an effect of performance-enhancing drugs, since tested and untested federations also differ in era, equipment, and selection. **Paired attempts.** Within a meet, the third squat attempt exceeds the opener by a median 15 kg (medians 160 $\rightarrow$ 180; $n = 10^5$ pairs; Wilcoxon signed-rank $p \rightarrow 0$), the paired-sample counterpart of the two-sample comparison above. **Equipment.** Dots differs across equipment categories (ANOVA $F = 5.3 \times 10^3$; Kruskal–Wallis agrees, $H = 1.2 \times 10^4$; all six Bonferroni post-hoc pairs reject), with multi-ply highest (median Dots 401 vs. 328 raw). **Extreme values.** A GEV fit to equal-size block maxima of recent totals gives shape $\xi = -0.02$ with bootstrap CI $[-0.20, 3.77]$: a population strength *ceiling is not supported*, since the CI includes zero. **Time trend.** Median Dots drifted down over 2000–2024 (Spearman $\rho = -0.54$); one possible explanation is broadening participation, but the trend is not by itself evidence of declining strength. **Sequential detection and power.** A Wald sequential probability ratio test on the chronological stream of just-below/above indicators accepts the bunching hypothesis after a stopping time of only $N = 27$ (Fig. 7). An a-priori calculation for the same design ($p = 0.6$ vs. 0.5) requires $n = 211$ for 0.90 power at $\alpha = 0.05$, and makes the Type I/II tradeoff concrete: at fixed $n = 100$, tightening α from 0.10 to 0.01 raises β from 0.23 to 0.63. **Independence.** After deduplicating to one row per lifter, the

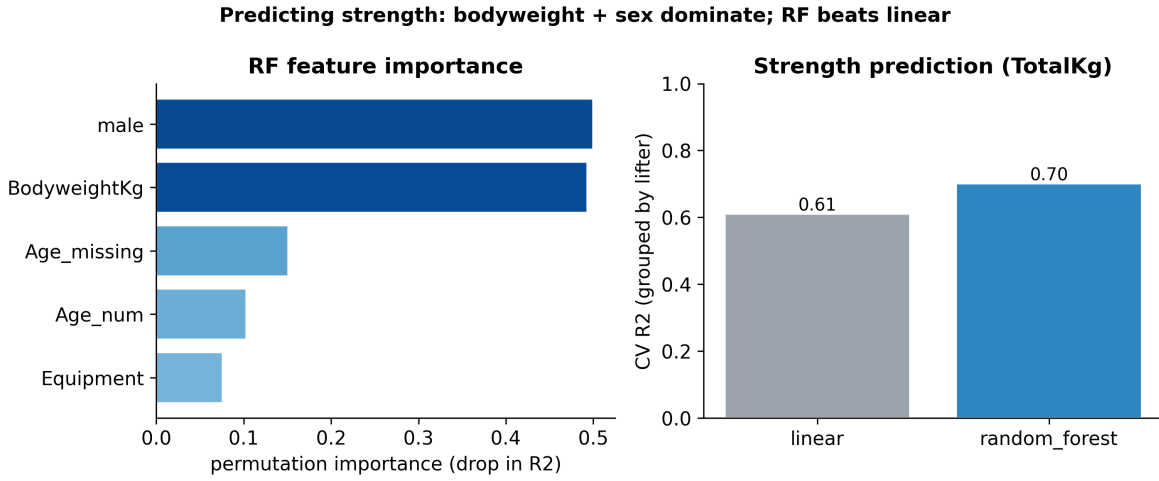


Fig. 5. H4: strength prediction; importance and grouped-CV R^2 .

association between tested status and cutting is negligible (Cramér's $V = 0.008$), illustrating how pseudo-replication inflates significance. **Residual normality.** The H3 (all-row fit) residuals are left-skewed with a heavy lower tail (skew -0.77 ; Fig. 8), so they are not exactly normal. This does not undermine the slope: at this n its sampling distribution is asymptotically normal under standard conditions, and we report HC3-robust standard errors. A formal normality test rejects even in modest subsamples ($n \approx 300$), as expected for genuinely non-normal residuals, so we treat it as a diagnostic and lead with effect sizes.

III. METHODS

Data and preprocessing. We use a pinned snapshot of OpenPowerlifting [9] (3,941,811 rows, public domain; the SHA-256 is recorded for exact reproducibility). The unit of analysis is the *attempt* for H1 (the mechanism is within-competitor, so the grid is robust to clustering; attempts do cluster within lifters, so the attempt-level CI is optimistic in width and the conclusion rests on the effect size, 96% on-grid vs. the 20% no-preference baseline) and one *row per lifter* for the cross-sectional tests; *de-heaping* drops exact round/half-kilogram weigh-ins so the bunching signal is not digit preference. For prediction, all cross-validation is grouped by lifter so the same athlete never appears in both train and test.

GLRT and the boundary (H1). With log-likelihood ℓ , the statistic is $2 \ln \Lambda = 2(\ell_1 - \ell_0)$. Under regularity $2 \ln \Lambda \rightarrow \chi^2$, but when the null fixes a parameter on the boundary of its space the limit becomes a mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ [7], [8]; we estimate the null distribution directly by simulating data under the fitted null model (parametric bootstrap) and comparing the observed statistic to the simulated quantiles. Concretely, for within-cell position $u \in \{0, \dots, 4\}$ on the 0.5 kg lattice the model is

$$P(u) = \pi \mathbf{1}[u = 0] + (1 - \pi) \frac{1}{5}, \quad \pi \in [0, 1], \quad (1)$$

and the GLRT contrasts $\pi = 0$ (no grid preference, on the boundary) with the constrained MLE $\hat{\pi} \geq 0$.

Density discontinuity (H2). For a cutoff c the manipulation test contrasts the one-sided densities, $\theta = \ln \hat{f}^+(c) - \ln \hat{f}^-(c)$, estimated by local polynomials without pre-binning [2], [3]; $\theta < 0$ ($t < 0$) indicates excess mass just below c . We complement the formal test with a de-heaped log count-ratio as the interpretable effect size, counting weigh-ins over the windows $(c - 0.5, c]$ and $(c, c + 0.5]$ kg.

Allometry (H3). OLS on log-log data gives the scaling exponent b ; we report HC3-robust standard errors (one row per lifter), a Wald statistic $W = (\hat{b} - 2/3)/\widehat{\text{se}}(\hat{b})$, and a pooled $\text{Sex} \times \log(\text{BW})$ interaction whose coefficient is exactly the men-minus-women slope difference. Correlations use the Fisher z -transform for confidence intervals.

Learning and the rest of the toolbox. Prediction uses linear and random-forest regression and logistic classification, evaluated by grouped cross-validation (R^2 , adjusted R^2 , RMSE, MAE, permutation importance, VIF, AUC/PR-AUC). Supporting analyses use a bootstrap mixture-LRT with AIC/BIC, a GEV extreme-value fit, a Wald sequential probability ratio test (with decision boundaries $A = \ln \frac{1-\beta}{\alpha}$, $B = \ln \frac{\beta}{1-\alpha}$, a stopping time, and an expected-sample-size approximation), one- and two-sample nonparametric tests (Wilcoxon signed-rank; Mann-Whitney), an F -test/Kruskal-Wallis omnibus with post-hoc comparisons, and a χ^2 independence test with Cramér's V and standardized residuals. The one-sided test of $b > 2/3$ for men ($z = 46.8$) is our Neyman-Pearson example: uniformly most powerful in the one-parameter exponential-family idealization, with the GLRT as its composite-hypothesis generalization. The extreme-value fit uses the GEV distribution

$$G(x) = \exp \left\{ - \left(1 + \xi \frac{x-\mu}{\sigma} \right)^{-1/\xi} \right\}, \quad (2)$$

where $\xi < 0$ implies a bounded upper tail (a strength ceiling) and $\xi \geq 0$ does not; we estimate ξ from equal-size block maxima and bootstrap its interval. The sequential test's

The two-component shape of raw Total tracks sex; Dots normalization removes it

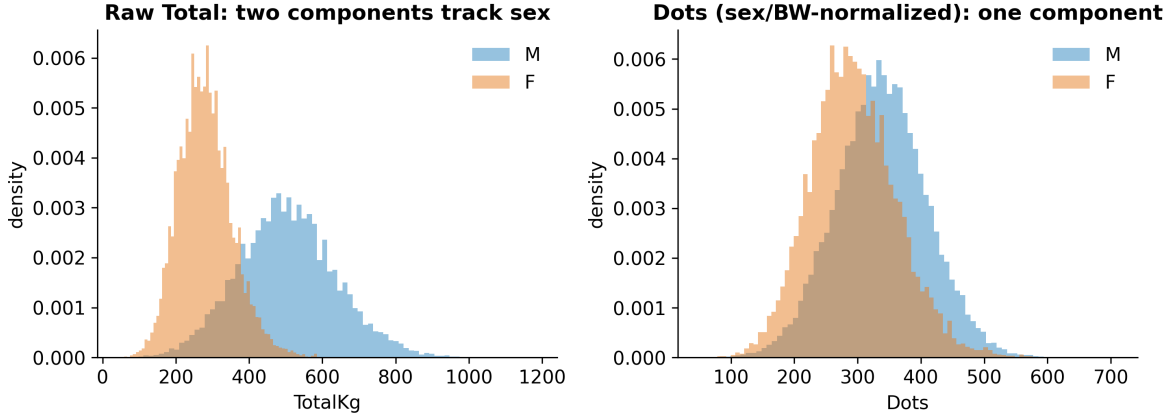


Fig. 6. The apparent “mixture” in raw Total is sex; Dots removes it.

Sequential test (stopping time) and a-priori power

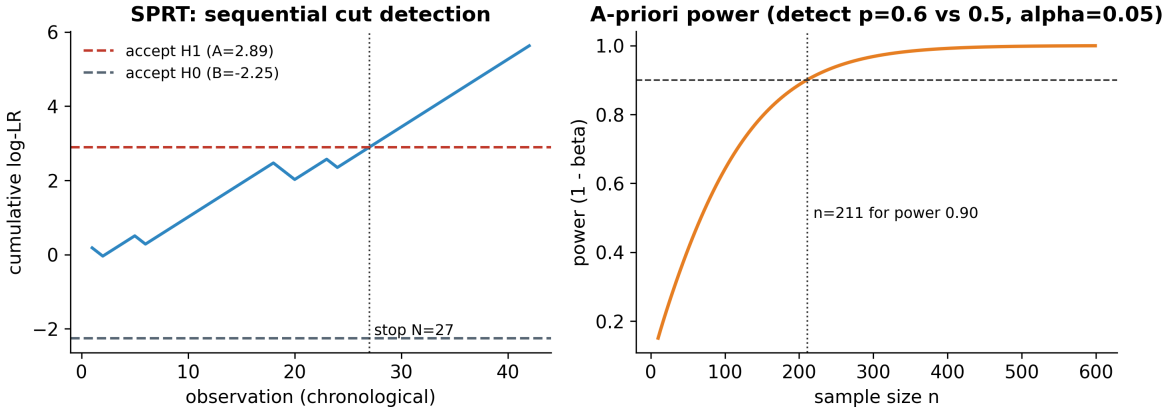


Fig. 7. Sequential test (stopping time $N = 27$) and a-priori power ($n = 211$ for power 0.90).

expected sample size under the alternative is approximated by $\mathbb{E}[N | H_1] \approx [(1 - \beta)A + \beta B] / [p_1 s_1 + (1 - p_1)s_0]$, with s_0, s_1 the per-observation log-likelihood increments. Table II maps the course toolbox onto the study.

Reproducibility. Every result is produced by a small set of scripts from the pinned snapshot; numbers are written to machine-readable files, figures are exported at 300 DPI, and an independent re-derivation from the raw data reproduced every headline value reported here. All analyses use Python 3.12 with numpy/scipy/statsmodels/scikit-learn and the rddensity implementation of the CJM test; exact versions are pinned in the repository.

IV. DISCUSSION

Conclusions. The two numbers a competitor controls leave measurable, falsification-surviving traces. Attempt loads are quantized to the plate grid; bodyweight bunches specifically below class limits, and not at controls or placebos in the bunching direction. Beyond strategy, the data also bear on the limits of strength: scaling with bodyweight is sex-specific and

departs from the isometric $2/3$, and basic meet-record variables already explain $\sim 70\%$ of the variance in the total. The same dataset supports the descriptive, testing, and prediction analyses.

Limitations and null results. This is an observational study: we identify population-level patterns consistent with weight cutting, not the intent of any individual. We report three null results as genuine findings. The 120 kg boundary (the entry to the unbounded 120+ class) does *not* show significant bunching after correction ($t = -1.6$, $p = 0.11$), consistent with weaker cutting pressure at that open-ended boundary; the extreme-value analysis does *not* support a strength ceiling (the ξ confidence interval includes zero); and we find no material association between tested status and cutting once pseudo-replication is removed. On the methods side, lifter identity is name-based, the per-lifter meet is sampled at random, and the formal bunching test is restricted to the modern kg-only era. The 83 kg signal is robust to the federation set: the de-heaped log-ratio is $+1.92$ across the IPF-affiliated federations

H3 residuals: a heavy lower tail (non-normal); the slope is CLT-robust at large n

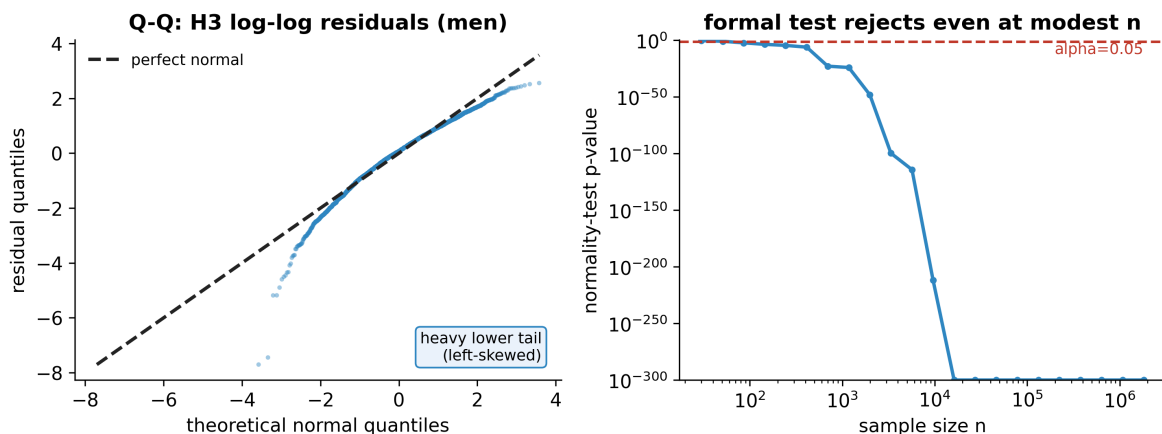


Fig. 8. H3 residuals: heavy lower tail; the slope is CLT-robust at large n .

TABLE II
COVERAGE OF THE COURSE TOOLBOX.

Concept	Where used
GLRT, MLE, χ^2 goodness-of-fit	H1 grid model
Boundary LRT, parametric bootstrap	H1 null calibration
Density discontinuity (McCrary/CJM)	H2 bunching
Multiple testing (Bonf./Holm/Šidák/BH)	H2 limits, H3
OLS, R^2 , Wald, interaction term	H3 allometry
One-/two-sided, one-sample tests	H1, H3
Neyman–Pearson MP/UMP framing	H3 one-sided test
Pearson/Spearman, Fisher CI	H3
Regression, random forest, logistic	H4 prediction
Learning metrics, grouped CV, VIF	H4
Mixture LRT, AIC/BIC	structure
Welch t / Mann–Whitney (two-sample)	tested vs. untested
Wilcoxon signed-rank (paired)	attempt 1 vs. 3
F /ANOVA, Kruskal–Wallis, post-hoc	equipment
χ^2 independence, Cramér’s V	tested $\times$ cut
Extreme-value theory (GEV)	strength ceiling
Sequential Wald (SPRT), stopping time	cut detection
Power, Type I/II tradeoff	a-priori power

used here and +1.66 when restricted to IPF/USAPL alone. Several extensions remain for future work: more federations and eras, a structural bunching model that recovers the share of manipulators, and a multiverse robustness check.

REFERENCES

- [1] H. J. Kleven, “Bunching,” *Annual Review of Economics*, vol. 8, pp. 435–464, 2016.
- [2] J. McCrary, “Manipulation of the running variable in the regression discontinuity design: A density test,” *J. Econometrics*, vol. 142, no. 2, pp. 698–714, 2008.
- [3] M. D. Cattaneo, M. Jansson, and X. Ma, “Simple local polynomial density estimators,” *J. Amer. Statist. Assoc.*, vol. 115, no. 531, pp. 1449–1455, 2020.
- [4] E. Franchini, C. J. Brito, and G. G. Artioli, “Weight loss in combat sports: physiological, psychological and performance effects,” *J. Int. Soc. Sports Nutrition*, vol. 9, art. 52, 2012.
- [5] J. H. J. Einmahl and J. R. Magnus, “Records in athletics through extreme-value theory,” *J. Amer. Statist. Assoc.*, vol. 103, no. 484, pp. 1382–1391, 2008.
- [6] J. C. Peyen, R. Nuermaimaiti, and J. Cunningham, “Discussing diminishing returns: A new scoring system for powerlifting,” arXiv:2503.13040, 2025.
- [7] H. Chernoff, “On the distribution of the likelihood ratio,” *Ann. Math. Statist.*, vol. 25, no. 3, pp. 573–578, 1954.
- [8] S. G. Self and K.-Y. Liang, “Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions,” *J. Amer. Statist. Assoc.*, vol. 82, no. 398, pp. 605–610, 1987.
- [9] OpenPowerlifting, <https://www.openpowerlifting.org> (data released into the public domain).